

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Answers:

1. Remote Rural Branch Office – Recommended Network Solution

Solution

Internet Connection:

- Use **4G/5G LTE** as the primary internet connection.
- Use **Satellite Internet (e.g., Starlink or VSAT)** as a backup for uninterrupted connectivity.

Networking Devices:

- Business-grade router with **dual-WAN** support.
- Managed Gigabit switch.
- Wi-Fi 6 wireless access point.
- UPS (Uninterruptible Power Supply) for power backup.

Security Mechanisms:

- Site-to-site **VPN** to securely connect with headquarters.
- Next-Generation Firewall (NGFW).
- WPA3 encryption for Wi-Fi.
- Multi-Factor Authentication (MFA) for remote access.

Justification

Requirement	Recommendation	Reason
Bandwidth	4G/5G Internet	Supports cloud applications and video conferencing with reasonable speed.
Reliability	Satellite backup	Ensures business continuity during mobile network failures.
Cost	LTE as primary	Lower installation and maintenance cost than leased lines.
Security	VPN + Firewall + MFA	Protects company data during transmission.
Future Scalability	Dual-WAN router and Wi-Fi 6	Easy to upgrade when better connectivity becomes available.

Conclusion

Using **4G/5G with satellite backup**, secured through VPN and firewall, provides a reliable, cost-effective, and scalable solution for the remote office.

2. University Campus Wireless Network Design

Solution

Access Point Placement

- Install APs in classrooms, libraries, laboratories, hostels, and common areas.
- Place APs centrally with minimal overlap.
- Perform a wireless site survey before installation.

Frequency Band Selection

- **5 GHz** for classrooms and laboratories (high speed, less interference).
- **2.4 GHz** for hostels and large coverage areas.
- Use dual-band Wi-Fi 6 access points.

Authentication Method

- WPA3-Enterprise.
- IEEE 802.1X authentication with a RADIUS server.
- Separate Guest Wi-Fi using VLAN.

Network Design

Component	Recommendation
Wireless Standard	Wi-Fi 6 (802.11ax)
Access Points	Centrally placed dual-band APs
Frequency	5 GHz + 2.4 GHz
Authentication	WPA3-Enterprise + RADIUS
Guest Network	Separate VLAN

Justification

Requirement	Solution
Coverage	Strategic AP placement reduces dead zones.
Performance	5 GHz minimizes congestion for heavy users.
Security	WPA3-Enterprise prevents unauthorized access.
Cost	Limited APs with proper placement reduce deployment cost.

Conclusion

A **Wi-Fi 6 dual-band network with WPA3-Enterprise authentication and optimized AP placement** provides excellent coverage, supports over 3,000 users, maintains security, and keeps costs under control.

3. Secure Network Architecture for Financial Institution

Proposed Network Architecture

Network Segmentation

- Use **VLANs** to separate:
 - Employee network
 - Customer database servers
 - Online banking servers
 - Guest network
- Control communication between VLANs using Access Control Lists (ACLs).

Firewall Deployment

- Deploy a **Next-Generation Firewall (NGFW)** at the network perimeter.
- Use an **Internal Firewall** to protect sensitive banking servers.

Authentication Methods

- Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC).
- Active Directory (AD) with centralized authentication.

Intrusion Detection

- Deploy **Intrusion Detection/Prevention System (IDS/IPS)**.
- Enable continuous security monitoring and logging.

Architecture Summary

Component	Recommendation
Segmentation	VLANs + ACLs
Firewall	NGFW + Internal Firewall
Authentication	MFA + RBAC + Active Directory
Threat Detection	IDS/IPS
Encryption	TLS/SSL and VPN

Justification

Requirement	Solution
Security	VLANs isolate sensitive systems and reduce attack spread.
Performance	VLANs reduce broadcast traffic and improve efficiency.
Compliance	MFA, RBAC, IDS/IPS, and encryption satisfy banking security requirements.
Budget	VLANs use existing switches, reducing infrastructure costs while improving security.

Conclusion

A secure architecture using **VLAN segmentation, NGFW, MFA, RBAC, IDS/IPS, and encrypted communication** provides strong protection for sensitive financial data while maintaining high performance, regulatory compliance, and cost efficiency.